Practical 7

Urine Analysis

7.1 Test for Sugar - Benedict's test

Benedict's test is used as a simple test for reducing sugars. A reducing sugar is a carbohydrate possessing either a free aldehyde or free ketone functional group as part of its molecular structure. This includes all monosaccharides (eg. glucose, fructose, galactose) and many disaccharides, including lactose and maltose.

Benedict's test is most commonly used to test for the presence of glucose in urine. Glucose found to be present in urine is an indication of Diabetes mellitus

Apparatus:

Benedict's solution (fresh; certainly not more than 3 months old), Dropper, Test-tube, Test-tube holder.

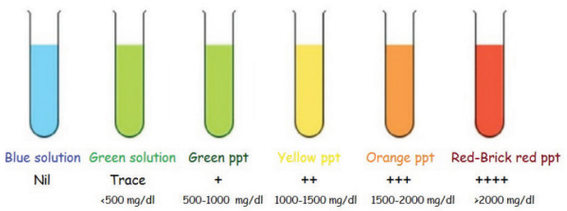
Quality checking of the Benedict's solution: Benedict's solution is blue in color. In order to check purity of Benedict's solution take 5 ml of Benedict's solution in test tube and heat it. If it does not change color, it means, it is pure.

Procedure:

- Take 5 ml (one teaspoon) of Benedict's solution in the test-tube.
- Holding the test-tube with the holder, heat it over a spirit lamp till the Benedict's Solution boils without overflowing.
- Drop 8 to 10 drops of urine into the boiling Benedict's solution.
- After again boiling the mixture, let it cool down.
- While cooling, the mixture changes color.
- Observe the color change and precipitate formation and analyze the test result

Result interpretation:

The colour of the mixture serves as a guide to the amount of sugar in the urine:



(ppt – precipitate)

Color	Approximate glucose mg/dl	Indication
Blue solution	Nil	
Green solution	<500 mg/dl	Trace
Green ppt	500-1000 mg/dl	+
Yellow ppt	1000-1500 mg/dl	++

Practical 7_295-296.indd 295 07/05/18 3:59 PM